

DARRELL J. BULLOCK

Staff Software Engineer | Technical Lead | Engineering Manager

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Software engineering leader with 20+ years of experience spanning individual contribution, technical architecture, and team leadership. Equally comfortable writing production code and guiding engineering organizations — from founding engineer building zero-to-one systems to staff-level technical authority owning architecture for 50+ production services. Deep expertise in backend engineering, LLM/AI integration, cloud infrastructure, and SRE practices. Consistent track record of measurable impact: reducing costs, improving reliability, and scaling team performance in EdTech and healthcare environments.

CORE COMPETENCIES

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| ▸ Technical Leadership & Org Scaling | ▸ System Architecture & Design Patterns | ▸ LLM / AI Integration & Optimization |
| ▸ Backend Engineering (Python, Ruby, Node.js) | ▸ Frontend Development (React, Next.js, TS) | ▸ Cloud Infrastructure (AWS) |
| ▸ PostgreSQL & Database Performance | ▸ Observability & SRE Practices | ▸ CI/CD & DevOps |
| ▸ Docker & Kubernetes | ▸ GraphQL & REST API Design | ▸ Microservices Architecture |
| ▸ React Native (iOS & Android) | ▸ HIPAA Compliance & Healthcare IT | ▸ Mentorship & Career Development |
| ▸ Agile / Scrum Methodologies | ▸ ADA / WCAG Accessibility | ▸ EdTech & Curriculum Development |

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Currus, Inc.

Jul 2025 – Dec 2025

Technical architecture lead and hands-on engineer for a data-driven financial services platform, establishing observability and reliability patterns and scaling engineering practices to production-grade SRE standards.

- Architected scalable microservices (FastAPI) with comprehensive observability stack — logging, metrics, distributed tracing — achieving 99.9% uptime and reducing mean time to recovery from 45 min to under 8 min.
- Optimized high-volume data pipelines processing 2M+ financial records/day by refactoring PostgreSQL queries and implementing async message-queue patterns, cutting processing time 60%.
- Led chapter-style engineering team via weekly 1-on-1s with 5 engineers and code-review standards, reducing production incidents by 40%.
- Built complex, high-performance React interfaces using advanced patterns — custom hooks, context-based state management, memoization, and code-splitting — to surface real-time pipeline metrics and pharmacy data at scale.
- Reduced AWS infrastructure costs 25% by implementing SQS-based async workflows while improving job reliability.
- Leveraged AWS services including EC2, S3, RDS, Lambda, ECS, and CloudWatch to deploy and operate containerized workloads with automated scaling and monitoring.

Technologies: React, FastAPI, Python, PostgreSQL, Apache Airflow, Docker, AWS (SQS, CloudWatch)

Technology Lead | Code Differently

Jan 2025 – Jul 2025

Led technical instruction and mentorship for a K-12/college EdTech program, designing full-stack curriculum and coaching 30+ students from beginner to job-ready.

- Scaled student capabilities to job-ready level with React + NodeJS + Java/Spring Boot curriculum including deploying applications to AWS (EC2, S3, RDS); 85% of graduates secured internships or entry-level roles.
- Built inclusive learning environment for multilingual and diverse learners; improved engagement scores 40% through scaffolded instruction.
- Mentored 30+ students through 1-on-1 coaching on system design, code quality, and professional collaboration; 90% reported increased confidence.

Technologies: React, NodeJS, Java, Spring Boot, EdTech curriculum design, inclusive teaching methodologies

Software Engineering Contractor | Thomas Jefferson University Hospitals

Jul 2023 – Jul 2024

Technical lead building zero-to-one healthcare applications supporting 15,000+ physicians and staff, with emphasis on reliability, observability, and cost optimization in HIPAA-regulated environments.

- Built respirator mask fit-test application from zero to production (15,000+ users) with 99.95% uptime using Python/Django, Node.js/Express, PostgreSQL, React/Next.js, TypeScript.
- Reduced API infrastructure costs 35% through efficient GraphQL schema design, AWS SQS async processing, and query refactoring — lowering monthly AWS spend from \$12K to \$7.8K.
- Scaled real-time clinical FastAPI microservices handling 50K+ daily requests at sub-200ms latency via AWS SQS message queuing.
- Architected event-driven serverless workflows with AWS Lambda, using single-purpose functions triggered by SQS, S3, and API Gateway events, with cold-start optimization and IAM least-privilege scoping.
- Leveraged additional AWS services including EC2, S3, RDS, ECS, API Gateway, and CloudWatch for containerized deployment, storage, and observability.
- Modernized legacy ColdFusion and Ruby on Rails systems to React/TypeScript/Python stack, reducing incident response time 55%.
- Built React Native iOS/Android apps for clinical workflows serving 2,500+ mobile users with offline-first architecture.
- Mentored 3 junior developers on architectural patterns and code quality standards, reducing production bugs 45%.

Technologies: Python (Django, FastAPI, Flask), Node.js, TypeScript, NestJS, React, Next.js, PostgreSQL, MS SQL Server, Azure, React Native, AWS (SQS, Lambda)

Senior Backend Engineer | Mathison.io

Mar 2022 – Mar 2023

Backend engineer for a SaaS Equity Index platform measuring employer DEI efforts, focusing on API architecture, data pipeline engineering, and production reliability.

- Architected Ruby on Rails / GraphQL APIs handling 1M+ daily requests at 99.9% uptime, reducing average response time from 850ms to 180ms.
- Built zero-to-one data pipeline processing 500K+ employer records/month with Domo/PostgreSQL integration; reduced data-sync failures 90% and enabled real-time analytics for 200+ enterprise customers.
- Designed role-based authorization system from scratch (Authzed) supporting 10K+ users across multi-tenant SaaS architecture.
- Reduced PostgreSQL query costs 40% via schema redesign and connection pooling; improved dashboard load times from 4.5s to 1.2s.
- Maintained production stability through on-call rotation and post-mortems, reducing P1 incidents 65%.

Technologies: Ruby on Rails, GraphQL, PostgreSQL, Domo, Authzed

Technical Lead, Software Engineering | Einstein Healthcare Network

Jul 1998 – Mar 2022

Primary technical authority and chapter lead over a 24-year tenure, owning architectural decisions for 50+ production services, scaling engineering teams, and delivering mission-critical healthcare applications in HIPAA-compliant environments.

Leadership & Engineering Organization

- Scaled engineering team from 3 to 15+ engineers via chapter-style leadership, 400+ performance reviews, and mentoring engineers from junior to senior level; maintained <5% attrition.

- Built SRE culture achieving 99.95% uptime for mission-critical systems by implementing observability stack (logging, metrics, distributed tracing, alerting) and SLOs across 20+ applications.
- Served as architectural authority for data models, API boundaries, microservices design, and infrastructure decisions across 50+ interconnected healthcare applications.
- Reduced infrastructure spend 45% over 3 years (\$180K/yr) by migrating to containerized microservices, implementing auto-scaling, and rightsizing AWS resources.

Hands-On Engineering & Zero-to-One Systems

- Built COVID-19 employee tracker from zero to production serving 15,000+ users (React, Next.js, TypeScript, Node.js/NestJS, MS SQL Server, MongoDB, Python), processing 100K+ test results with 24/7 availability.
- Architected containerized FastAPI microservices handling 200K+ async jobs/month with Docker/Kubernetes and AWS SQS (retry logic + dead-letter queues), achieving 99.9% message delivery.
- Optimized PostgreSQL and SQL Server reducing query latency 70% via schema redesign and Django ORM refactoring; enabled real-time reporting for 10K+ concurrent users.
- Built AWS infrastructure (EC2, Lambda, S3, DynamoDB, SQS) supporting 24/7 healthcare operations with auto-scaling, reducing cloud spend 35%.

Code Quality, Observability & Engineering Excellence

- Reduced mean time to detection from 25 min to under 3 min via structured logging, application metrics, distributed tracing, and real-time CloudWatch alerting.
- Implemented CI/CD pipelines with 5,000+ automated tests per release cycle, cutting production incidents 60% through quality gates and coverage thresholds.
- Led legacy ColdFusion-to-React/TypeScript/Python modernization saving \$200K/yr and improving uptime from 94% to 99.95%.
- Processed 5M+ healthcare records/year via SSIS pipelines with HIPAA compliance and 99.99% data accuracy.

Technologies: React, Next.js, TypeScript, Node.js, NestJS, Python (Django, FastAPI, Flask), Ruby on Rails, PostgreSQL, MS SQL Server, MongoDB, DynamoDB, React Native, Docker, Kubernetes, AWS (EC2, Lambda, S3, SQS, CloudWatch), GraphQL

EDUCATION

Bachelor of Science, Electrical Engineering | [Howard University, Washington, DC](#)

ADDITIONAL

Currently expanding expertise in: Elixir / Phoenix

Domain experience: Healthcare IT (HIPAA), EdTech, Financial Services, DEI / Equity platforms